When applying F_i^+, the space V_i gets replaced by K. Furthermore, let

$$\psi : K \to \bigoplus_{j \to i} V_j.$$

After applying F_i^-, K gets replaced by

$$K' = \left(\bigoplus_{j \to i} V_j \right) / (\operatorname{Im} \psi).$$

But

$$\operatorname{Im} \psi = K$$

and therefore

$$K' = \left(\bigoplus_{j \to i} V_j \right) / \left(\ker(\varphi : \bigoplus_{j \to i} V_j \to V_i) \right) = \operatorname{Im}(\varphi : \bigoplus_{j \to i} V_j \to V_i)$$

by the homomorphism theorem. Since φ was assumed to be surjective, we get

$$K' = V_i.$$

□

Proposition 6.6.7. *Let Q be a quiver, and let V be an indecomposable representation of Q. Then $F_i^+ V$ and $F_i^- V$ (whenever defined) are either indecomposable or 0.*

Proof. We prove the proposition for $F_i^+ V$; the case $F_i^- V$ follows similarly. By Proposition 6.6.5 it follows that either

$$\varphi : \bigoplus_{j \to i} V_j \to V_i$$

is surjective or $\dim V_i = 1, \dim V_j = 0, j \neq i$. In the last case

$$F_i^+ V = 0.$$

So we can assume that φ is surjective. In this case, assume that $F_i^+ V$ is decomposable as

$$F_i^+ V = X \oplus Y$$